Unlearning Results: NPO vs SCRUB on TOFU

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TL;DR

Both NPO and SCRUB stopped the model from reciting the forgotten facts, but both overshot: they pushed the forget data far past where a model that never saw it would be. NPO is the more balanced of the two; we have not yet shown that either one truly removed the knowledge.

- **Forgetting worked on the surface.** ROUGE-L on forget questions fell from 0.90 (original) to 0.24 (NPO) and 0.09 (SCRUB).
- **Both overshot.** Forget loss should end near the retrained model's 2.97. NPO ended at 5.65 and SCRUB at 29.87. An over-forgotten model is itself a sign that the data was once there.
- **Retention cost was small, NPO smaller.** Held-out retain ROUGE-L went from 0.94 to 0.89 (NPO) and 0.84 (SCRUB).
- **Neither gives a natural wrong answer.** The retrained model invents a plausible name; NPO goes off-topic and SCRUB puts a book-title-like string where the name should be.
- **Not yet tested:** relearning attack, paraphrased questions, and more than one seed.

Setup

We taught a small model some fake facts, then tried to make it forget a subset of them without breaking everything else. Run on a free Colab T4, one seed.

1. **Model:** `Qwen2.5-0.5B`, fine-tuned in full fp32.
2. **Data (TOFU, fictional authors):**
   - Forget set: `forget01`, 40 Q&As about 2 authors (Basil Mahfouz Al-Kuwaiti, Nikolai Abilov).
   - Retain set: 300 Q&As from `retain99`. 200 are used during unlearning, 100 are held out for evaluation.
   - General set: 25 `real_authors` + 25 `world_facts` questions, to check real-world knowledge.
3. **Stage 0 (learn):** 5 epochs on forget + retain. Forget loss fell from 2.15 to 0.05, so the model memorised the facts. This is the **original** model.
4. **Stage 1 (unlearn):** 30 steps from the original, lr 1e-5, batch 4.

- **NPO** (Zhang et al. 2024): a bounded version of gradient ascent on the forget set, plus normal training on retain data ($\beta = 0.1$, $\alpha = 1$).
- **SCRUB** (Kurmanji et al. 2023): the student is pushed away from the original model on forget data for the first 20 steps, and kept close to it on retain data every step.

5. **Retrained (gold standard):** the base model fine-tuned on the 300 retain Q&As only. It never saw the forget set, so it shows what perfect unlearning looks like.

Metrics:

- **Loss:** how surprised the model is by the true answer. Higher on forget = more forgotten.
- **ROUGE-L recall:** how much of the true answer appears in what the model actually generates. Lower on forget = more forgotten; retain and general should stay near the original.

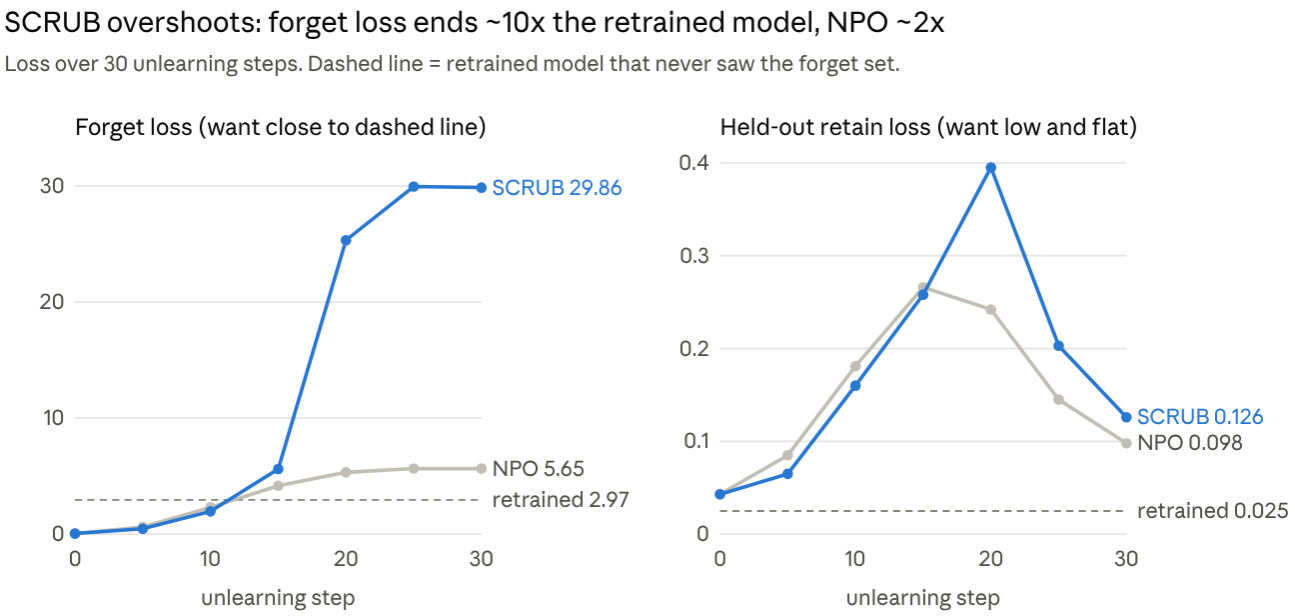
Results

NPO lands closest to the retrained model on forget ROUGE and keeps more retain performance; SCRUB forgets hardest but overshoots the most.

Model	Forget loss (target ≈ retrained)	Forget ROUGE-L (lower = forgot)	Held-out retain loss	Retain ROUGE-L	General ROUGE-L
Original	0.051	0.901	0.043	0.944	0.343
NPO	5.651	0.243	0.098	0.887	0.260
SCRUB	29.866	0.088	0.126	0.839	0.427
Retrained (gold)	2.967	0.368	0.025	0.978	0.397

Probe question (from the forget set): *What is the full name of the author born in Kuwait City, Kuwait on 08/09/1956?* True answer: Basil Mahfouz Al-Kuwaiti.

Model	Answer
Original	… is Basil Mahfouz Al-Kuwaiti. (correct)
NPO	The full name of the fictitious city on the African continent is "The African Metamorphosis".
SCRUB	… born in Kuwait City, Kuwait on the 8th of September, 1956 is "The Hussein Metamorphosis".
Retrained	… born in Kuwait City, Kuwait on August 09, 1956 is Laila Amira al-Faisal.



Notebook training logs, logged every 5 steps · retrained values from the evaluation table

Both methods pass the retrained forget loss between steps 10 and 15, then keep going. SCRUB jumps once its push-away phase runs to step 20; retain loss peaks there and recovers afterwards.

What the results mean

The goal of unlearning is to look like the retrained model, not to push forget loss as high as possible. Judged that way, NPO is closer on every forget metric, and SCRUB overshoots badly.

1. Forgetting happened, but only measured on the exact questions

Both methods stopped reciting the forget answers: forget ROUGE-L fell from 0.90 to 0.24 (NPO) and 0.09 (SCRUB). This shows the model no longer *says* the facts. It does not show the facts are gone from the weights; that needs the relearning attack (see Caveats).

2. Both methods over-forgot

The retrained model, which never saw these authors, has forget loss 2.97 and forget ROUGE-L 0.37. Both unlearned models go well past it:

- NPO: forget loss 5.65, about 2× the retrained value.
- SCRUB: forget loss 29.87, about 10×. The model assigns near-zero probability to the true answers, far less than to any random fact it never learned.

This matters because an unnaturally high loss on specific data is detectable. An attacker comparing losses could tell these 40 questions were once in training and were removed, which defeats part of the purpose. The table's "forget loss ↑" label in the notebook is misleading for this reason; the target is "≈ retrained".

3. Retention cost was small, and temporary during training

Held-out retain ROUGE-L dropped from 0.94 to 0.89 (NPO) and 0.84 (SCRUB). Retain loss rose more in relative terms (0.043 → 0.098 and 0.126) but stays low in absolute terms. During training, retain loss peaked (NPO 0.27 at step 15, SCRUB 0.40 at step 20) and then recovered once the forget push weakened. For SCRUB, step 20 is exactly when its "push away" phase stops.

4. The general-knowledge numbers are not reliable yet

General ROUGE-L moved around (0.26 to 0.43) with no clear pattern, and SCRUB even scored above the original. Two reasons to discount it:

- It is only 50 questions, so one or two answers move the average a lot.
- The original model is already damaged by Stage 0. Asked who India's first prime minister was, it answered "a young boy named Pratibha Choudhury", while NPO and SCRUB said Jawaharlal Nehru. Fine-tuning on fake authors made it invent fake people for real questions. We never measured the untouched base model, so we cannot say how much general knowledge any step cost.

5. The wrong answers look wrong in different ways

Ideal unlearning gives a fluent, plausible wrong answer, as the retrained model does ("Laila Amira al-Faisal"). Neither method does this on the probe:

- NPO drifts off-topic, answering about “a fictitious city on the African continent”.
- SCRUB keeps the sentence template but slots in a book-title-like phrase, “The Hussein Metamorphosis”, instead of a person’s name.

Both outputs are recognisably “unlearned” to a human reader. This is one question, so treat it as an example, not a measurement.

Caveats

These are first-run demo numbers; do not quote them as a method ranking yet.

- **Relearning attack not run.** The cell is in the notebook but has no output. Without it we cannot tell “removed” from “hidden”.
- **No paraphrase test.** All forget metrics use the exact training questions.
- **One seed, 40 forget examples.** Differences of a few ROUGE points are within noise.
- **Unequal compute.** SCRUB makes two updates per step for its first 20 steps (push-away + stay-close), so it gets 50 updates to NPO’s 30.
- **No base-model row.** The untouched `Qwen2.5-0.5B` was not evaluated, so general-knowledge loss from Stage 0 is not measured.
- **ROUGE-L recall rewards long answers.** A rambling answer can score well by chance; spot-check the raw outputs.
- **Interactive tests need the question format.** Typing a statement (e.g. “X was born in Y”) just makes every model agree with it; that tells us nothing.

Next steps

The priority is to find out whether the knowledge is gone, then to stop the overshoot.

- [ ] Run the relearning attack cell (already in the notebook) and compare each curve with the retrained one.
- [ ] Write ~10 paraphrased forget questions and score all models on them.
- [ ] Stop at the retrained level instead of a fixed 30 steps. NPO passed forget loss 2.97 between steps 10 and 15; SCRUB passed it between 10 and 15 as well. Try early stopping on forget loss, or fewer SCRUB push-away steps.
- [ ] Add the untouched base model as a row in the evaluation table.
- [ ] Repeat with 3 seeds and report mean ± spread.
- [ ] Match compute: give NPO 50 updates, or count SCRUB’s updates the same way.
- [ ] Third method: SalUn’s weight-saliency mask wrapped around NPO.